

Module 4: Lexical Semantics & Embeddings

WordNet Synsets, TF-IDF, Positive PMI, Word2Vec (Skip-Gram SGNS) & GloVe Log-Bilinear Model

Module IV: Lexical Semantics, Vector Space Models & Distributed Word Embeddings

1. Lexical Semantics: Word Senses, Synonyms, Antonyms, Hypernyms & Hyponyms
2. WordNet Ontology Architecture: Synsets, Taxonomic Distance & Similarity Metrics
3. Distributional Hypothesis: "You shall know a word by the company it keeps"
4. Term-Document Matrix & Term Frequency-Inverse Document Frequency (TF-IDF)
5. Term-Context Matrices & Pointwise Mutual Information (PMI) / Positive PMI
6. Dimensionality Reduction in Semantics: Latent Semantic Analysis (LSA / SVD)
7. Word2Vec Architecture (Mikolov et al., 2013): Continuous Bag-of-Words (CBOW)
8. Word2Vec Skip-Gram Model with Negative Sampling (SGNS) Mathematical Objective
9. Vector Arithmetic & Analogical Reasoning ($\mathbf{v}_{\text{King}} - \mathbf{v}_{\text{Man}} + \mathbf{v}_{\text{Woman}} \approx \mathbf{v}_{\text{Queen}}$)
10. GloVe (Global Vectors for Word Representation): Log-Bilinear Matrix Factorization
11. FastText: Subword N-Gram Embeddings for Morphology & Out-of-Vocabulary (OOV)
12. Comprehensive Solved BIT Mesra & GATE Exam Question Bank (8 Questions)

Topic 4 & 5: Vector Space Models (TF-IDF & Positive PMI)

1. Term Frequency-Inverse Document Frequency (TF-IDF):

$$\text{TF-IDF}(t, d, D) = \text{TF}(t, d) \times \text{IDF}(t, D)$$

$$\text{TF}(t, d) = \frac{f_{t,d}}{\sum_{t' \in d} f_{t',d}}, \quad \text{IDF}(t, D) = \log_{10} \left(\frac{|D|}{|\{d \in D : t \in d\}|} \right)$$

2. Positive Pointwise Mutual Information (PPMI):

$$\text{PMI}(w, c) = \log_2 \left(\frac{P(w, c)}{P(w)P(c)} \right) = \log_2 \left(\frac{C(w, c) \cdot N}{C(w) \cdot C(c)} \right)$$

$$\text{PPMI}(w, c) = \max(\text{PMI}(w, c), 0)$$

Topic 7 & 8: Word2Vec (Skip-Gram with Negative Sampling – SGNS)

Word2Vec learns dense low-dimensional continuous vector embeddings ($d \approx 300$) where semantically similar words are mapped to proximate points in vector space.

Skip-Gram with Negative Sampling (SGNS) Loss Function

For a target word w with true context word c_{pos} and k randomly drawn negative words $c_{\text{neg},1} \dots c_{\text{neg},k}$ from noise distribution $P_n(w) \propto f(w)^{3/4}$:

$$\mathcal{L}_{\text{SGNS}} = \log \sigma(\mathbf{v}_{c_{\text{pos}}}^T \mathbf{v}_w) + \sum_{i=1}^k \mathbb{E}_{c_{\text{neg},i} \sim P_n} \left[\log \sigma(-\mathbf{v}_{c_{\text{neg},i}}^T \mathbf{v}_w) \right]$$

Where $\sigma(z) = \frac{1}{1+e^{-z}}$ is the logistic sigmoid function. Maximizes dot product with actual context words while pushing away negative noise samples!

Topic 10: GloVe (Global Vectors for Word Representation)

GloVe Log-Bilinear Objective Function (Pennington, Socher, Manning, 2014):

$$J = \sum_{i,j=1}^V f(X_{ij}) \left(\mathbf{w}_i^T \tilde{\mathbf{w}}_j + b_i + \tilde{b}_j - \log X_{ij} \right)^2$$

Where X_{ij} is the co-occurrence count of word i with context word j , and $f(X_{ij}) = \min(1, (X_{ij}/x_{\max})^\alpha)$ is a weighting function that caps the influence of extremely frequent stop words (with $\alpha = 0.75, x_{\max} = 100$).

Top BIT Mesra Exam Questions & Answers (Module IV)

Q1. Compare CBOW and Skip-Gram models in Word2Vec across 4 architectural parameters. (8 Marks)

1. **Prediction Task:** CBOW predicts the target center word given a window of surrounding context words; Skip-Gram predicts the surrounding context words given the center word.
2. **Training Speed:** CBOW trains several times faster than Skip-Gram because context word vectors are averaged in a single forward pass.
3. **Performance on Rare Words:** Skip-Gram is substantially superior for rare words and small datasets because each target-context pair is treated as a separate training sample.
4. **Vector Representations:** CBOW smooths over context distribution; Skip-Gram preserves multi-sense representations.